

VERIFICATION REPORT

PARTHENOPE → PSYCHE → THEMIS*Low-thrust trajectory chain — independent physics & feasibility audit*

Independent physics and feasibility audit of `optimal_asteroid_paths/pk1/ppt_lt_chain_v2.pk1`, generated by `Python_Consolidated/verify_ppt_lt_chain_full.py`. Every value below is independently re-derived from raw SPICE states and the saved Sims-Flanagan throttle profiles. The verifier does not trust the saved numbers; it recomputes them from scratch and reports any mismatch.

Headline Result*All checks pass.*

Item	Value
Triplet	PARTHENOPE → PSYCHE → THEMIS
Architecture	Direct (no flyby)
Launch Δv (impulsive, excluded from objective)	7.0000 km/s (constraint ≤ 7.0)
Post-launch Δv (objective)	13.1053 km/s
Mission duration	16.794 yr (constraint ≤ 30)
LT-chain initial mass	1500 kg
Final delivered mass	974.70 kg (65.0% of LT start)
Spacecraft launch mass	3000 kg
All Sims-Flanagan legs converged	Yes
Total checks pass / fail	146 / 0

Mission Timeline

Event	UTC	Elapsed (yr)
Earth launch	2029 JUL 28 21:06:34	0.00
Arrive PARTHENOPE	2033 APR 29 17:12:43	3.75
Depart PARTHENOPE	2033 NOV 10 11:56:40	4.29
Arrive PSYCHE	2040 OCT 09 09:02:25	11.20
Depart PSYCHE	2041 OCT 09 15:02:21	12.20
Arrive THEMIS	2046 MAY 15 00:40:56	16.79

Δv Breakdown

All values independently re-derived. Heliocentric ECLIPJ2000 frame, units km/s.

Burn	Δv	Type	In objective?
Earth launch (C3 / v_∞)	7.0000	Impulsive (launcher)	No
Earth → PARTHENOPE	6.1541	Integrated LT	Yes
PARTHENOPE → PSYCHE	2.4801	Integrated LT	Yes
PSYCHE → THEMIS	4.4711	Integrated LT	Yes
Total post-launch (objective)	13.1053	LT sum	—

Per-Leg Detailed Audit

Leg 1 — Earth → PARTHENOPE

Quantity	Value
Time of flight (TOF)	3.7531 years (118,440,369 s)
Lambert m-revs (initial guess for LT solver)	1
Mass in / out	1500.0000 → 1225.1115 kg (propellant burned: 274.8885 kg)
Integrated Δv (saved)	6.154096 km/s
Integrated Δv (re-integrated)	6.154096 km/s
Mismatch	0.000000e+00 km/s (< 1e-3 required)
Endpoint position error vs target asteroid	23.6776 km
Endpoint velocity error vs target asteroid	6.6397e-06 km/s
Sims-Flanagan converged	True

Thrust profile (15 segments × 3 components)

- Max |throttle vector| (engine constraint ≤ 1): 0.480290
- Mean thrust: 71.15 mN
- Peak thrust: 144.09 mN
- Duty cycle (>5% thrust): 100.0%

Seg	Time (yr)	u_x	u_y	u_z	$ u $	Thrust (N)
0	0.1251	-0.1775	-0.2800	+0.0225	0.3323	0.0997
1	0.3753	-0.0653	-0.1883	+0.0130	0.1998	0.0599
2	0.6255	-0.1171	-0.1224	+0.0155	0.1701	0.0510

Seg	Time (yr)	u_x	u_y	u_z	u	Thrust (N)
3	0.8757	-0.1624	-0.1188	+0.0196	0.2022	0.0607
4	1.1259	-0.1711	-0.1516	+0.0222	0.2297	0.0689
5	1.3762	-0.1386	-0.1876	+0.0217	0.2342	0.0703
6	1.6264	-0.0692	-0.1963	+0.0167	0.2088	0.0626
7	1.8766	+0.0083	-0.1479	+0.0075	0.1483	0.0445
8	2.1268	+0.0275	-0.0618	-0.0015	0.0677	0.0203
9	2.3770	-0.0340	+0.0638	-0.0050	0.0724	0.0217
10	2.6272	-0.1142	+0.1185	+0.0001	0.1645	0.0494
11	2.8774	-0.2373	+0.1178	+0.0100	0.2651	0.0795
12	3.1276	-0.3504	+0.0509	+0.0221	0.3548	0.1064
13	3.3778	-0.4214	-0.0632	+0.0333	0.4274	0.1282
14	3.6280	-0.4349	-0.1996	+0.0414	0.4803	0.1441

Leg 2 — PARTHENOPE → PSYCHE

Quantity	Value
Time of flight (TOF)	6.9127 years (218,149,546 s)
Lambert m-revs (initial guess for LT solver)	1
Mass in / out	1225.1115 → 1129.1351 kg (propellant burned: 95.9764 kg)
Integrated Δv (saved)	2.480079 km/s
Integrated Δv (re-integrated)	2.480079 km/s
Mismatch	0.000000e+00 km/s (< 1e-3 required)
Endpoint position error vs target asteroid	2.6836 km
Endpoint velocity error vs target asteroid	6.9600e-07 km/s
Sims-Flanagan converged	True

Thrust profile (15 segments × 3 components)

- Max |throttle vector| (engine constraint ≤ 1): 0.096386
- Mean thrust: 13.42 mN
- Peak thrust: 28.92 mN
- Duty cycle (>5% thrust): 33.3%

Seg	Time (yr)	u_x	u_y	u_z	u	Thrust (N)
0	0.2304	-0.0161	-0.0001	-0.0035	0.0165	0.0049

Seg	Time (yr)	u_x	u_y	u_z	u	Thrust (N)
1	0.6913	-0.0186	+0.0102	+0.0173	0.0273	0.0082
2	1.1521	-0.0239	+0.0075	+0.0281	0.0376	0.0113
3	1.6130	-0.0186	+0.0057	+0.0225	0.0298	0.0089
4	2.0738	-0.0233	+0.0211	+0.0000	0.0315	0.0094
5	2.5347	-0.0497	+0.0183	-0.0190	0.0562	0.0169
6	2.9955	-0.0615	-0.0100	-0.0246	0.0670	0.0201
7	3.4564	-0.0454	-0.0340	-0.0182	0.0595	0.0179
8	3.9172	-0.0192	-0.0370	-0.0038	0.0419	0.0126
9	4.3781	+0.0005	-0.0241	+0.0135	0.0276	0.0083
10	4.8389	+0.0103	-0.0055	+0.0265	0.0289	0.0087
11	5.2998	+0.0142	+0.0145	+0.0290	0.0354	0.0106
12	5.7606	+0.0084	+0.0410	+0.0189	0.0459	0.0138
13	6.2215	-0.0228	+0.0656	+0.0002	0.0695	0.0208
14	6.6823	-0.0748	+0.0583	-0.0172	0.0964	0.0289

Leg 3 — PSYCHE → THEMIS

Quantity	Value
Time of flight (TOF)	4.5952 years (145,013,915 s)
Lambert m-revs (initial guess for LT solver)	0
Mass in / out	1129.1351 → 974.7037 kg (propellant burned: 154.4314 kg)
Integrated Δv (saved)	4.471129 km/s
Integrated Δv (re-integrated)	4.471129 km/s
Mismatch	0.000000e+00 km/s (< 1e-3 required)
Endpoint position error vs target asteroid	8.1186 km
Endpoint velocity error vs target asteroid	2.0690e-06 km/s
Sims-Flanagan converged	True

Thrust profile (15 segments × 3 components)

- Max |throttle vector| (engine constraint ≤ 1): 0.199427
- Mean thrust: 32.55 mN
- Peak thrust: 59.83 mN
- Duty cycle (>5% thrust): 100.0%

Seg	Time (yr)	u_x	u_y	u_z	u	Thrust (N)
0	0.1532	+0.1295	-0.0449	-0.0484	0.1454	0.0436
1	0.4595	+0.0918	-0.0604	-0.0457	0.1190	0.0357
2	0.7659	+0.0737	-0.0765	-0.0373	0.1126	0.0338
3	1.0722	+0.0685	-0.0828	-0.0251	0.1103	0.0331
4	1.3786	+0.0658	-0.0770	-0.0105	0.1018	0.0306
5	1.6849	+0.0557	-0.0641	+0.0058	0.0852	0.0256
6	1.9913	+0.0338	-0.0536	+0.0228	0.0674	0.0202
7	2.2976	+0.0035	-0.0526	+0.0374	0.0646	0.0194
8	2.6040	-0.0218	-0.0612	+0.0457	0.0794	0.0238
9	2.9103	-0.0360	-0.0733	+0.0485	0.0949	0.0285
10	3.2166	-0.0435	-0.0797	+0.0461	0.1018	0.0306
11	3.5230	-0.0543	-0.0764	+0.0389	0.1015	0.0304
12	3.8293	-0.0786	-0.0668	+0.0274	0.1067	0.0320
13	4.1357	-0.1216	-0.0626	+0.0130	0.1374	0.0412
14	4.4420	-0.1822	-0.0811	-0.0026	0.1994	0.0598

All Checks

Detailed pass/fail results, organized by verification section.

1. Data Integrity

82 pass / 0 fail

Check	Status	Detail
Top-level pkI is a dict	✓	
Key 'best_ordering' present	✓	
Key 'best_flyby' present	✓	
Key 'verified' present	✓	
Key 'config' present	✓	
Key 'surrogate' present	✓	
Triplet has 3 names	✓	['PARTHENOPE', 'PSYCHE', 'THEMIS']
best_flyby is None (direct architecture)	✓	flyby = None
verified['epochs'] present	✓	
verified['verified_legs'] present	✓	
verified['launch_dv_kms'] present	✓	
verified['post_launch_dv_kms_full'] present	✓	
verified['m_final_kg_full'] present	✓	
verified['m_revs'] present	✓	
verified['feasibility'] present	✓	
Epoch 'et_launch' present	✓	
Epoch 'et_a1_arr' present	✓	
Epoch 'et_a1_dep' present	✓	
Epoch 'et_a2_arr' present	✓	
Epoch 'et_a2_dep' present	✓	
Epoch 'et_a3_arr' present	✓	
Exactly 3 verified LT legs (E→A1, A1→A2, A2→A3)	✓	Got 3
Leg 0 — key label	✓	
Leg 0 — key et_start	✓	
Leg 0 — key et_end	✓	

Check	Status	Detail
Leg 0 — key tof_yr	✓	
Leg 0 — key m_revs	✓	
Leg 0 — key m_in_kg	✓	
Leg 0 — key m_out_kg	✓	
Leg 0 — key converged	✓	
Leg 0 — key dv_integral_kms	✓	
Leg 0 — key thrust_profile	✓	
Leg 0 — key pos_err_km	✓	
Leg 0 — key vel_err_kms	✓	
Leg 0 — thrust_profile.throttle_unit_vector	✓	
Leg 0 — thrust_profile.thrust_magnitude_N	✓	
Leg 0 — thrust_profile.thrust_max_N	✓	
Leg 0 — thrust_profile.time_yr_from_leg_start	✓	
Leg 0 — thrust_profile.segment_dt_yr	✓	
Leg 0 — throttle shape (15, 3)	✓	Got (15, 3)
Leg 0 — throttles all finite	✓	
Leg 0 — m_in / m_out finite	✓	
Leg 1 — key label	✓	
Leg 1 — key et_start	✓	
Leg 1 — key et_end	✓	
Leg 1 — key tof_yr	✓	
Leg 1 — key m_revs	✓	
Leg 1 — key m_in_kg	✓	
Leg 1 — key m_out_kg	✓	
Leg 1 — key converged	✓	
Leg 1 — key dv_integral_kms	✓	
Leg 1 — key thrust_profile	✓	
Leg 1 — key pos_err_km	✓	
Leg 1 — key vel_err_kms	✓	
Leg 1 — thrust_profile.throttle_unit_vector	✓	

Check	Status	Detail
Leg 1 — thrust_profile.thrust_magnitude_N	✓	
Leg 1 — thrust_profile.thrust_max_N	✓	
Leg 1 — thrust_profile.time_yr_from_leg_start	✓	
Leg 1 — thrust_profile.segment_dt_yr	✓	
Leg 1 — throttle shape (15, 3)	✓	Got (15, 3)
Leg 1 — throttles all finite	✓	
Leg 1 — m_in / m_out finite	✓	
Leg 2 — key label	✓	
Leg 2 — key et_start	✓	
Leg 2 — key et_end	✓	
Leg 2 — key tof_yr	✓	
Leg 2 — key m_revs	✓	
Leg 2 — key m_in_kg	✓	
Leg 2 — key m_out_kg	✓	
Leg 2 — key converged	✓	
Leg 2 — key dv_integral_kms	✓	
Leg 2 — key thrust_profile	✓	
Leg 2 — key pos_err_km	✓	
Leg 2 — key vel_err_kms	✓	
Leg 2 — thrust_profile.throttle_unit_vector	✓	
Leg 2 — thrust_profile.thrust_magnitude_N	✓	
Leg 2 — thrust_profile.thrust_max_N	✓	
Leg 2 — thrust_profile.time_yr_from_leg_start	✓	
Leg 2 — thrust_profile.segment_dt_yr	✓	
Leg 2 — throttle shape (15, 3)	✓	Got (15, 3)
Leg 2 — throttles all finite	✓	
Leg 2 — m_in / m_out finite	✓	

2. Constraint Compliance

From **CONSTRAINTS_AND_OUTPUTS.md** — 17 pass / 0 fail

Check	Status	Detail
Launch $\Delta v \leq 7.0$ km/s	✓	7.0000 km/s
Mission duration ≤ 30.0 yr	✓	16.794 yr
Stay 1 ≥ 3.0 mo	✓	6.40 mo
Stay 2 ≥ 3.0 mo	✓	12.00 mo
Spacecraft launch mass = 3000.0 kg	✓	3000.0
LT chain initial mass = 1500.0 kg	✓	1500.0
Isp = 3100.0 s	✓	3100.0
Thrust cap = 0.3 N	✓	0.3
Leg 0 $m_{in} ==$ LT chain initial mass	✓	1500.0000 vs 1500.0
Leg 0 — max throttle ≤ 1.001 (engine constraint)	✓	max u = 0.4803
Leg 1 — max throttle ≤ 1.001 (engine constraint)	✓	max u = 0.0964
Leg 2 — max throttle ≤ 1.001 (engine constraint)	✓	max u = 0.1994
Mass strictly decreases each leg	✓	['1500.0', '1225.1', '1129.1', '974.7']
Final mass > 0	✓	974.70 kg
Final mass equals saved $m_{final_kg_full}$	✓	Last leg $m_{out} = 974.70$, saved = 974.70
No flyby in this trajectory	✓	flyby_name = None
All LT legs converged (per saved flag)	✓	

3. Physics Re-derivation

Independent of saved values — 5 pass / 0 fail

Check	Status	Detail
Lambert E→A1 (m=1) converges	✓	Exit flag = 1
Re-derived launch Δv matches saved within 1 mm/s	✓	Recomputed = 7.000001, saved = 7.000001
Lambert PARTHENOPE→PSYCHE (m=1) converges	✓	Exit flag = 1
Lambert PSYCHE→THEMIS (m=0) converges	✓	Exit flag = 1
Saved post_launch_dv == sum of per-leg dv_integral	✓	Sum = 13.105304, saved = 13.105304

4. LT Integration Accuracy

Re-integrate each leg — 18 pass / 0 fail

Check	Status	Detail
Leg 0 (Earth→A1) — re-integrated pos err near saved	✓	Recomputed = 23.68 km, saved = 23.68 km
Leg 0 (Earth→A1) — re-integrated vel err near saved	✓	Recomputed = 6.64e-06 km/s, saved = 6.64e-06
Leg 0 — re-integrated dv_integral matches saved	✓	Recomputed = 6.154096, saved = 6.154096
Leg 0 — re-integrated m_out matches saved	✓	Recomputed = 1225.1115, saved = 1225.1115 kg
Leg 0 — m_out < m_in (mass decreases)	✓	in = 1500.00, out = 1225.11 kg
Leg 0 — m_out matches Tsiolkovsky from dv	✓	Tsiolkovsky 1225.1115 vs integrator 1225.1115
Leg 1 (A1→A2) — re-integrated pos err near saved	✓	Recomputed = 2.68 km, saved = 2.68 km
Leg 1 (A1→A2) — re-integrated vel err near saved	✓	Recomputed = 6.96e-07 km/s, saved = 6.96e-07
Leg 1 — re-integrated dv_integral matches saved	✓	Recomputed = 2.480079, saved = 2.480079
Leg 1 — re-integrated m_out matches saved	✓	Recomputed = 1129.1351, saved = 1129.1351 kg
Leg 1 — m_out < m_in (mass decreases)	✓	in = 1225.11, out = 1129.14 kg
Leg 1 — m_out matches Tsiolkovsky from dv	✓	Tsiolkovsky 1129.1351 vs integrator 1129.1351
Leg 2 (A2→A3) — re-integrated pos err near saved	✓	Recomputed = 8.12 km, saved = 8.12 km
Leg 2 (A2→A3) — re-integrated vel err near saved	✓	Recomputed = 2.07e-06 km/s, saved = 2.07e-06
Leg 2 — re-integrated dv_integral matches saved	✓	Recomputed = 4.471129, saved = 4.471129
Leg 2 — re-integrated m_out matches saved	✓	Recomputed = 974.7037, saved = 974.7037 kg
Leg 2 — m_out < m_in (mass decreases)	✓	in = 1129.14, out = 974.70 kg
Leg 2 — m_out matches Tsiolkovsky from dv	✓	Tsiolkovsky 974.7037 vs integrator 974.7037

5. Numerical Health

15 pass / 0 fail

Check	Status	Detail
All integrated values finite (no NaN/Inf)	✓	
Leg 0 — heliocentric distance in [0.5, 5.0] AU	✓	min=1.015, max=2.674 AU
Leg 1 — heliocentric distance in [0.5, 5.0] AU	✓	min=2.179, max=2.816 AU
Leg 2 — heliocentric distance in [0.5, 5.0] AU	✓	min=3.035, max=3.247 AU
Mass strictly monotonic non-increasing per segment	✓	Across 48 segments
Max throttle across all segments ≤ 1.0001	✓	max u globally = 0.480290
Leg 0 — per-segment dv $\leq$ thrust ceiling	✓	max dv/ceiling = 0.4803
Leg 1 — per-segment dv $\leq$ thrust ceiling	✓	max dv/ceiling = 0.0964
Leg 2 — per-segment dv $\leq$ thrust ceiling	✓	max dv/ceiling = 0.1994
Leg 0 — sum of per-segment dvs == integral	✓	sum = 6.1540962463, integral = 6.1540962463
Leg 1 — sum of per-segment dvs == integral	✓	sum = 2.4800787203, integral = 2.4800787203
Leg 2 — sum of per-segment dvs == integral	✓	sum = 4.4711294422, integral = 4.4711294422
Leg 0 — trajectory samples within sensible bounds	✓	(sample bounds checked above)
Leg 1 — trajectory samples within sensible bounds	✓	(sample bounds checked above)
Leg 2 — trajectory samples within sensible bounds	✓	(sample bounds checked above)

6. SPICE / BSP Coverage

9 pass / 0 fail

Check	Status	Detail
Asteroid PARTHENOPE (SPICE ID 20000011) state retrievable at launch	✓	
Asteroid PSYCHE (SPICE ID 20000016) state retrievable at launch	✓	
Asteroid THEMIS (SPICE ID 20000024) state retrievable at launch	✓	
Launch (2029 JUL 28 21:06:34) within BSP coverage	✓	
Arrive A1 (2033 APR 29 17:12:43) within BSP coverage	✓	
Depart A1 (2033 NOV 10 11:56:40) within BSP coverage	✓	
Arrive A2 (2040 OCT 09 09:02:25) within BSP coverage	✓	
Depart A2 (2041 OCT 09 15:02:21) within BSP coverage	✓	

Check	Status	Detail
Arrive A3 (2046 MAY 15 00:40:56) within BSP coverage	✓	

Methodology Notes

Independent re-derivation

Every Δv , mass, and trajectory endpoint reported above is independently re-derived by the verifier:

- **Launch Δv** = $|V1_lambert - v_Earth|$ from a fresh Lambert solve at the saved `et_launch` and `et_a1_arr`, with the `m_revs[0] = 1` revolution branch the solver chose.
- **Per-leg LT trajectory** = forward Sims-Flanagan integration using the saved 15-segment throttle profile, starting from the pre-leg state (Lambert V1 at Earth for the launch leg, asteroid co-orbit velocity for asteroid-to-asteroid legs). The integrator is bit-identical to `lowthrust._forward_propagate` (same Kepler propagation, same impulsive kick at segment midpoint, same Tsiolkovsky mass update).
- **Endpoint check** = compare integrated final state to the target asteroid's SPICE state at `et_end`. The re-derived position error must match the solver's saved `pos_err_km` (within 5×) and the solver's saved `vel_err_kms` (within 5×). Both bounds are met.

Sources of accepted numerical residuals

- **Position errors at asteroid arrival** (3–24 km across legs): inherent residual of Sims-Flanagan with finite segment count (15 per leg). Tightening would require more segments or a collocation method.
- **Throttle saturation** (segments at $|u| = 1.000$): the engine is running at the 0.30 N max during those segments. This is expected and within the constraint.

What this verifier does NOT check

- Power-system mass model (assumes thrust is always available)
- Sun-pointing constraints on the solar array
- Operational margins beyond the hard physical limits
- Trajectory observability / navigation-uncertainty propagation
- Second-order perturbations (J2, n-body) — pure two-body around Sun
- Asteroid arrival trajectory targeting beyond rendezvous-velocity match